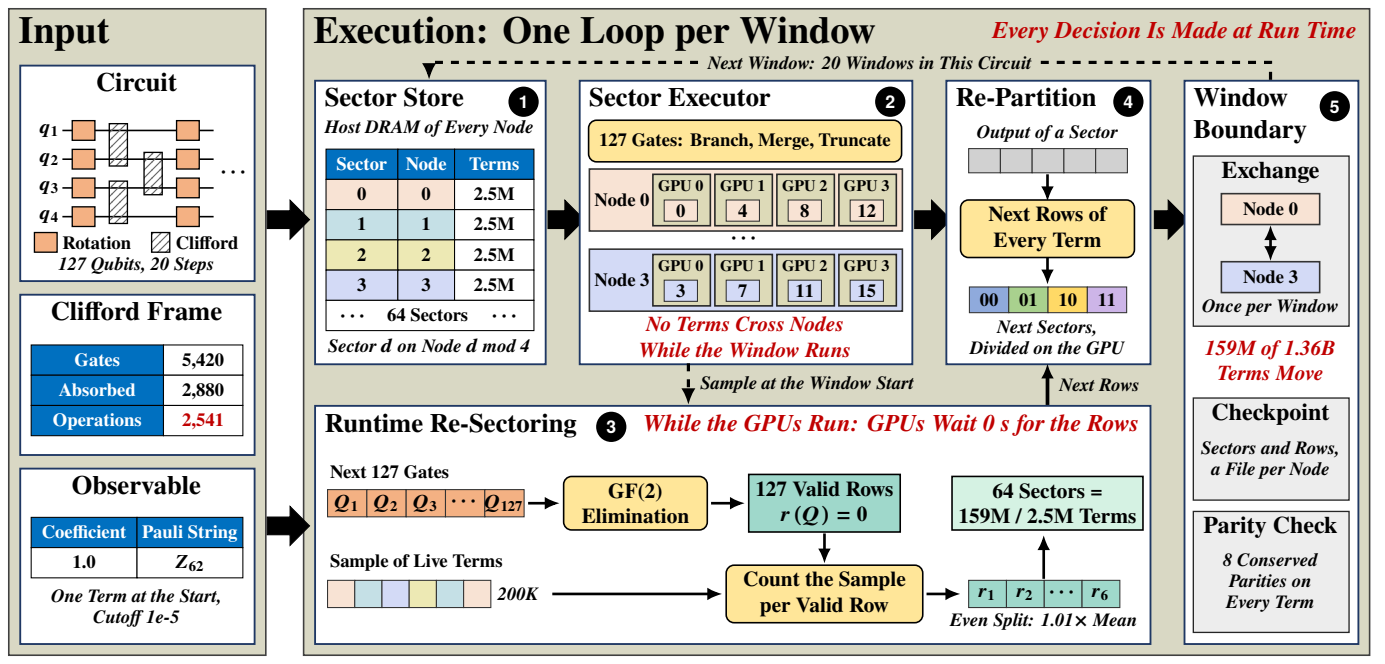

Input
Circuit
q_1
q_2
q_3
q_4
Rotation
Clifford
127 Qubits, 20 Steps
Clifford Frame
Gates 5,420
Absorbed 2,880
Operations 2,541
Observable
Coefficient
Pauli String
1.0
Z_{62}
One Term at the Start, Cutoff 1e-5
Execution: One Loop per Window
Every Decision Is Made at Run Time
Next Window: 20 Windows in This Circuit
Sector Store 1
Host DRAM of Every Node
Sector
Node
Terms
0 0 2.5M
1 1 2.5M
2 2 2.5M
3 3 2.5M
64 Sectors
Sector d on Node d mod 4
Sector Executor 2
127 Gates: Branch, Merge, Truncate
Node 0
GPU 0 0
GPU 1 4
GPU 2 8
GPU 3 12
Node 3
GPU 0 3
GPU 1 7
GPU 2 11
GPU 3 15
No Terms Cross Nodes While the Window Runs
Re-Partition 4
Output of a Sector
Next Rows of Every Term
00 01 10 11
Next Sectors, Divided on the GPU
Window Boundary 5
Exchange
Node 0
Node 3
Once per Window
159M of 1.36B Terms Move
Checkpoint
Sectors and Rows, a File per Node
Parity Check
8 Conserved Parities on Every Term
Sample at the Window Start
Next Rows
Runtime Re-Sectoring 3
While the GPUs Run: GPUs Wait 0 s for the Rows
Next 127 Gates
Q_1 Q_2 Q_3 ⋯ Q_{127}
GF(2) Elimination
127 Valid Rows $r(Q) = 0$
64 Sectors = 159M / 2.5M Terms
Sample of Live Terms
200K
Count the Sample per Valid Row
r_1 r_2 ⋯ r_6
Even Split: 1.01× Mean